ANALOG ELECTRONICS / CURRENT CONTROL

VOLTAGE-CONTROLLED CURRENT SINK

Design of a Multi-Range Programmable Active Load

TECHNICAL DESIGN REPORT

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Design tools	LTspice • Altium Designer
Project focus	Analog current control • Active-load design • PCB layout

This report documents the circuit architecture, design iterations, component selection, LTspice simulations, and Altium schematic/PCB implementation.

Abstract

This project concerns the design of a programmable voltage-controlled current sink intended to act as an active load during the electrical characterization and validation of integrated circuits and other electronic products. The design is organized around two complementary current-control approaches: an operational-amplifier/MOSFET linear current-sink stage for lower current ranges, and a boost-converter-based stage with inductor-current feedback for the high-current range. A microcontroller provides the programmable reference and manages range selection, while analog feedback circuits regulate the current.

The linear stage uses an OP07 operational amplifier, a pass MOSFET, and selectable current-sense resistors. Design iterations address MOSFET operating-region constraints, the influence of relay contact resistance on low-resistance sensing, and the need to separate the highest current range from the linear stage. The high-current stage uses a boost power topology, a 10 mΩ current-sense element, differential amplification, an integrator-based controller, a 100 kHz triangular carrier, PWM comparison, and a MOSFET gate driver. The design is simulated in LTspice, and the schematic and two-layer PCB are documented as having been laid out in Altium Designer.

The available project record documents circuit design, simulation, component selection, and PCB layout. It does not include bench measurement tables, calibration data, thermal test results, or a hardware demonstration; therefore, no measured performance or completed hardware validation is claimed in this report.

Keywords: Voltage-controlled current sink, VCCS, active load, current feedback, boost converter, PWM, STM32F103RCT6, LTspice, Altium Designer.

Executive Summary

- Purpose: provide a controllable current-sinking load for testing ICs under defined electrical load conditions.
- Control concept: convert a programmable voltage reference into a regulated current using negative feedback and a selected current-sense resistor.
- Low-/mid-current implementation: OP07-based feedback with a MOSFET pass element and selectable Rsense branches.
- High-current implementation: a boost-converter power stage regulated through inductor-current sensing and PWM control.
- Digital control: STM32F103RCT6 provides the DAC reference and controls range selection; USB-to-UART is specified for host communication.
- Design implementation: functional schematic blocks and a double-sided PCB are documented in Altium Designer; circuit behavior is explored in LTspice.

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1. Project Overview and Design Requirements

1.1 Project Purpose

The objective is to develop a voltage-programmable current sink that can operate as an active load when characterizing or validating integrated circuits. Instead of relying on a passive resistor whose current depends directly on the device voltage, the active load draws a commanded current and regulates that current through a feedback loop.

The project is specifically framed as a current-sink application: current is drawn from the device under test (DUT). The term VCCS is used in the project title, while the implemented functional description is a Voltage-Controlled Current Sink.

1.2 Design Requirements

Requirement	Design interpretation
Current programmability	Set the load current through a voltage reference generated by the controller.
Wide operating range	Cover low-current testing and a high-current operating region, with the overall design target extending to 10 A.
Load adaptability	Maintain current regulation under different DUT/load conditions within the available voltage compliance.
Accuracy and stability	Use current sensing and negative feedback to reduce the effect of device and load variation.
Configurable ranges	Select different current-sense resistors and control paths to improve resolution across ranges.
Monitoring and control	Provide a microcontroller interface for setpoint generation and range control.

1.3 Electrical Context

The design notes identify an available dual supply of +15 V and −15 V for analog circuitry. The microcontroller operates from a 3.3 V rail, while the analog multiplexer is supplied from 5 V. A 12 V rail is used for relay coils, and the gate-driver stage is described as operating from 15 V. These rails serve different functional blocks and should be checked against the final schematic and component datasheets.

> **Scope of this report**
> The report describes the documented design and simulation work. It does not assert that the PCB was fabricated, that the complete circuit was assembled, or that measured current accuracy was demonstrated.

2. System Architecture

The architecture separates current regulation into two power paths. The continuous linear current-sink path is used for the lower and intermediate ranges. A separate boost-based path is assigned to the highest-current range after the linear implementation was found to have insufficient voltage-compliance margin under the demanding low-supply/high-current condition.

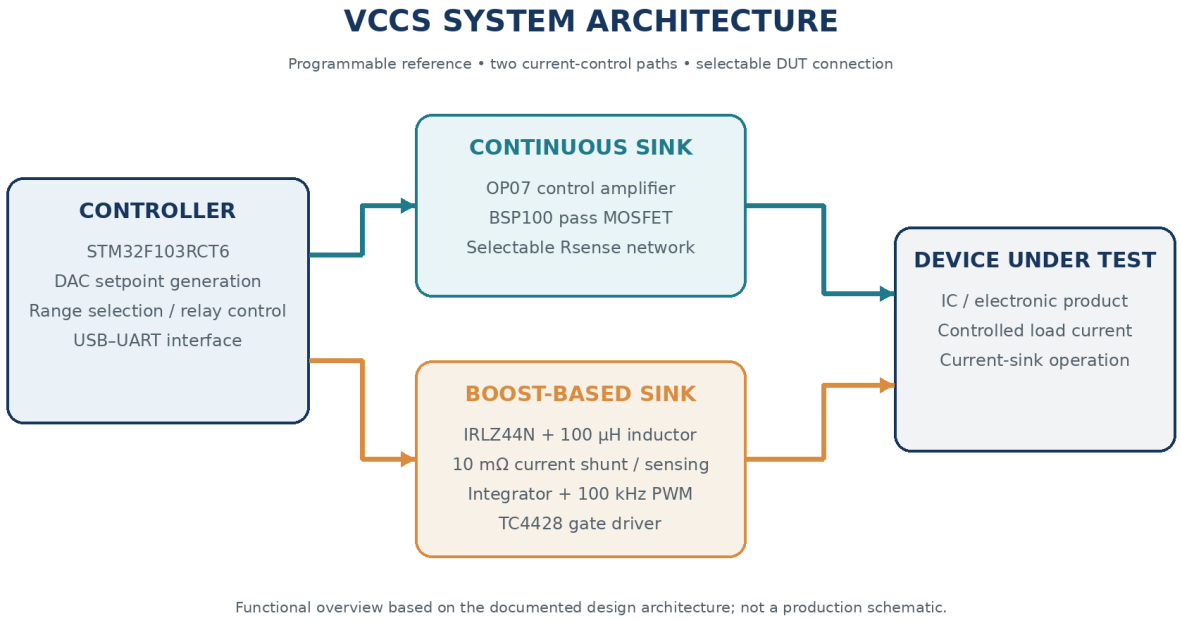


Figure 1. Functional overview of the two current-sink paths and the controller.

2.1 Functional Blocks

Block	Function
Controller	STM32F103RCT6 generates the analog current reference through its DAC and manages range selection and relay control. A USB-to-UART interface is specified for communication with a host computer.
Linear current-sink path	An OP07 error amplifier drives a MOSFET pass element. The current is determined by the reference voltage and the selected Rsense network.
Boost-based current-sink path	A boost power stage is controlled from inductor-current feedback. The feedback chain includes a shunt, differential amplifier, integrator, PWM comparator, and gate driver.
Range-selection network	The design uses an analog multiplexer and semiconductor switches for selecting feedback/sense branches, with relays used to route between the linear and boost paths.
DUT interface and supplies	The DUT is connected to the selected current-sink path. External supply rails support the analog, digital, relay, and gate-drive sections.

2.2 Design Evolution

1. A basic op-amp/MOSFET current sink was selected as the starting topology.
2. Multiple Rsense values were introduced to support several current ranges.
3. Relay-based range switching was examined; contact resistance became significant when the sense resistance was very small.
4. The feedback measurement was moved to the two terminals of the sense resistor and a differential amplifier was introduced.
5. MOSFET/BJT semiconductor switches and a multiplexer were adopted for selecting the lower-range sense branches.
6. The highest-current range was separated into a boost-based current-control stage to address the compliance limitations of the linear topology.
7. The boost feedback loop was compensated by adding a zero to reduce overshoot and settling time.

3. Linear Current-Sink Stage

3.1 Operating Principle

The basic stage applies the input reference voltage Vin to the non-inverting input of an OP07. The amplifier drives the gate of an N-channel MOSFET so that negative feedback forces the sensed voltage across Rsense to follow the reference. Neglecting input bias current and assuming the amplifier remains within its operating limits, the current through the sense resistor is approximately:

$$I_{SINK} \approx V_{IN} / R_{SENSE}$$

Because the MOSFET gate draws negligible steady-state current, the current through the pass device is set by the sense-resistor feedback loop. The MOSFET drain is connected to the supply being loaded, so the circuit draws a controlled current from that supply, subject to the MOSFET's safe operating area and available voltage headroom.

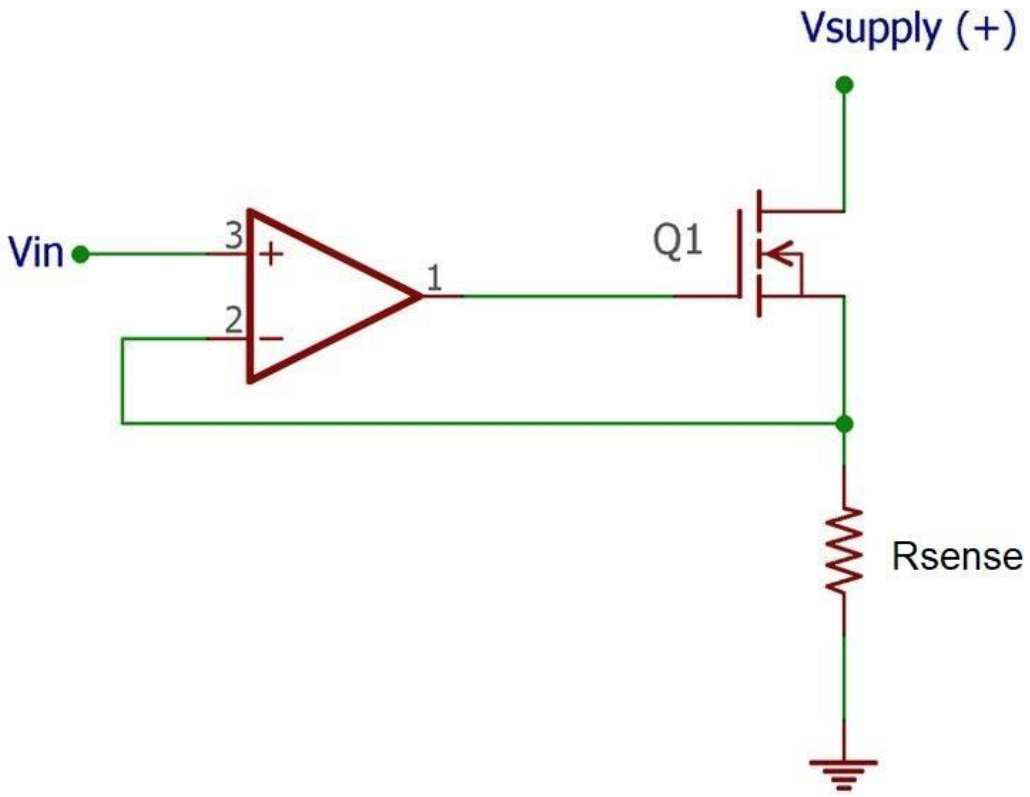


Figure 2. Basic op-amp/MOSFET voltage-controlled current-sink circuit.

3.2 Initial Current-Range Concept

The initial design divides the overall 0–10 A target into three ranges. The microcontroller DAC is specified to provide a 0–3.3 V reference. The initial sense-resistor and reference-voltage values are reproduced below as documented in the design record.

Intended current range	Rsense	DAC reference range
0–10 mA	330 Ω	0–3.3 V
10–250 mA	13.2 Ω	0.132–3.3 V
250 mA–10 A	0.1 Ω	0.25–1.0 V

The initial 10 A range created a difficult operating condition when the DUT supply was as low as 0.3 V. The source notes identify insufficient MOSFET voltage headroom and difficulty maintaining the required operating region. A reference-voltage divider and parallel MOSFETs were considered, but these changes did not provide adequate compliance margin.

3.3 Feedback Measurement and Range Switching

Relay contact resistance was initially considered negligible. However, with low-value Rsense elements, contact resistance becomes part of the measured path and can introduce a meaningful error. The design therefore moves the feedback measurement to the two terminals of Rsense and uses a differential amplifier. With matched resistor ratios, the documented difference-amplifier relationship is:

$$V_OUT = V_2 - V_1$$

A small resistor is also placed in the op-amp output path to limit the current delivered to the MOSFET gate. The output of the selected feedback network is routed through a 74HC4051D analog multiplexer controlled by the microcontroller.

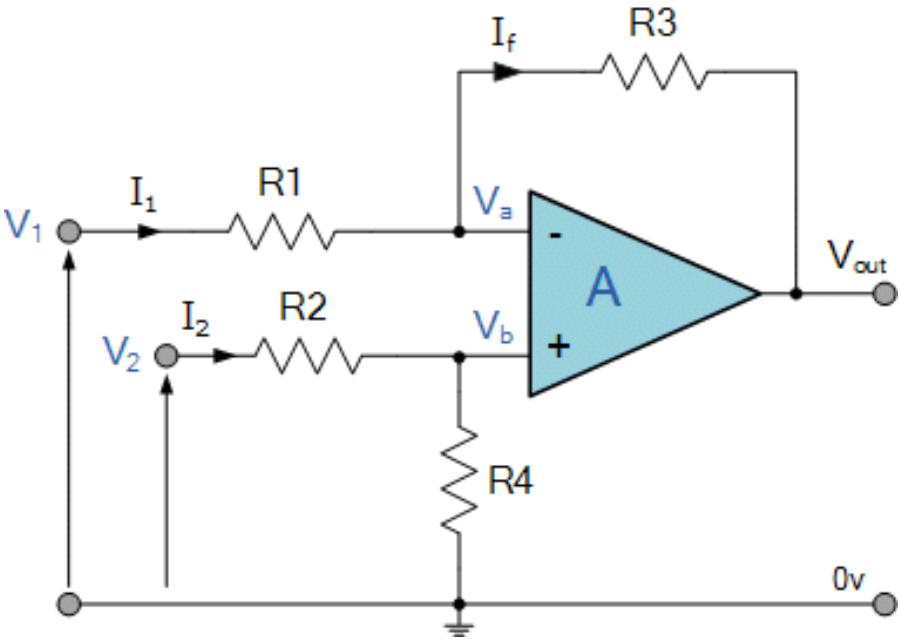


Figure 3. Differential-amplifier configuration used to sense the voltage across the current-sense resistor

The subsequent range-selection approach uses an IRF7303 MOSFET switch together with BC817 NPN transistors. The semiconductor switches connect or isolate the individual Rsense branches. This approach avoids the mechanical contact resistance and coil power associated with relays in the low-current sensing path.

3.4 Revised Linear-Stage Ranges

After the highest-current range was reassigned to a separate topology, the linear stage was documented as covering up to 1 A. The final table in the source design record is reproduced below.

Documented current range	Rsense	DAC reference range
0–10 mA	33 Ω	0–3.3 V
10–250 mA	1.2 Ω	0.12–3.0 V
250 mA–1 A	0.31 Ω	0.775–3.1 V

Parameter reconciliation required

The tabulated Rsense and voltage values do not directly reproduce all of the stated current-range labels under I = V/R if the listed voltage is interpreted as the voltage directly across Rsense. For example, 3.3 V across 33 Ω corresponds to 100 mA. An additional scaling factor, a table transcription issue, or a different reference definition may explain the discrepancy. The values should be reconciled against the final Altium schematic and circuit

calculations before fabrication or publication.

4. High-Current Boost-Based Stage

4.1 Motivation for a Separate Power Stage

The linear current-sink stage could not satisfy the documented high-current condition with adequate MOSFET voltage headroom. The design therefore assigns the 1–10 A region to a separate boost-converter-based stage. This separates the low-/mid-current precision path from the high-current power path.

4.2 Boost-Converter Operation

The boost converter stores energy in an inductor while the switching MOSFET is on. When the MOSFET turns off, the inductor current is transferred through the Schottky diode to the output capacitor and load. The capacitor reduces output-voltage ripple. In the ideal continuous-conduction model, the voltage conversion ratio is:

$$V_{OUT} = V_{IN} / (1 - D)$$

where D is the switch duty cycle. The inductor-current ripple is related to the input voltage, inductance, switching frequency, and duty cycle. The standard idealized expression used for design estimation is:

$$\Delta I_L \approx (V_{IN} \times D) / (L \times f_{SW})$$

The original design notes describe a high-current operating objective around 10 A and a low input-voltage region of approximately 0.3–1.9 V. The boost stage is then closed around the inductor current rather than operated as an open-loop converter.

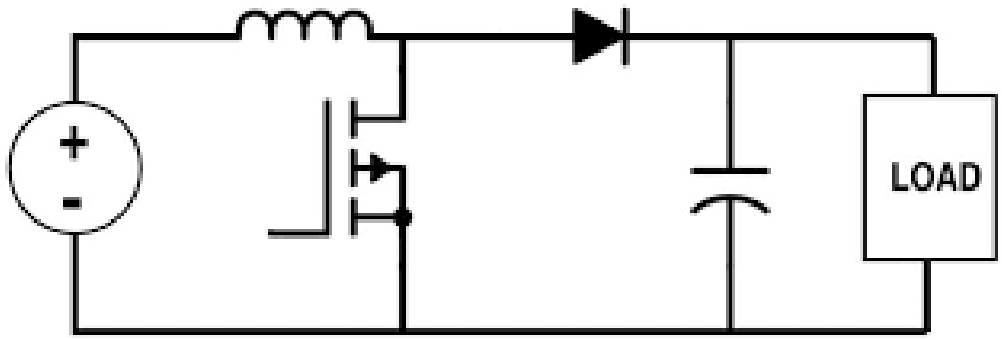


Figure 4. Simplified boost-converter power stage used as the basis for the high-current path

4.3 Current-Sensing Path

A 10 mΩ shunt is inserted in the inductor-current path. The voltage across the shunt is measured by a differential amplifier with a gain of 10. The sensed signal is scaled for comparison with the microcontroller-

derived reference. The control error indicates whether the actual current is below or above the commanded current.

$$V_SENSE,AMP = 10 \times (V_2 - V_1)$$

When the measured current is below the target, the controller increases the duty cycle. When the measured current exceeds the target, the duty cycle is reduced. An integrator converts the current error into a control voltage, which is compared with a triangular carrier to generate PWM.

4.4 PWM Generation and Gate Drive

The PWM carrier is specified at 100 kHz. A triangular waveform is compared with the integrator output. The comparator switches between its output states according to the relative levels of these signals, producing a PWM waveform whose duty cycle follows the control voltage. The design considers either a 3.3 V or 5 V triangular-wave amplitude.

The power MOSFET gate has a capacitive load. The design notes therefore place a TC4428CPA gate driver between the comparator output and the IRLZ44N gate to provide the required gate-charge current and switching capability.

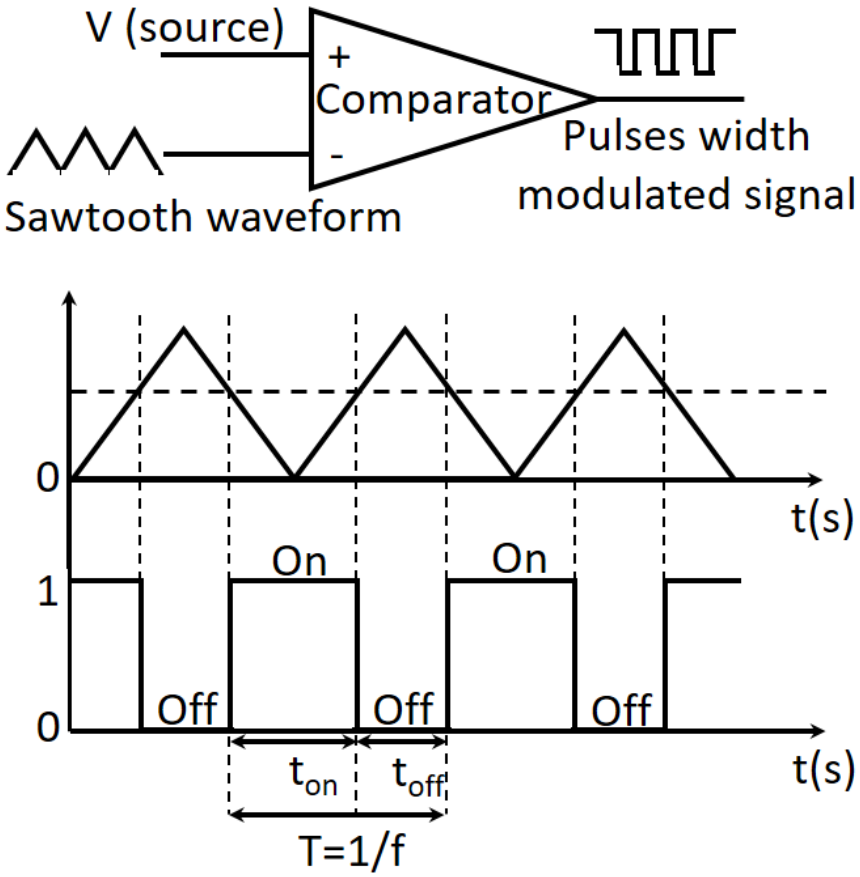


Figure 5. Triangular-carrier comparison and resulting PWM waveform.

4.5 Compensation and Transient Response

Initial LTspice results were described as relatively slow and exhibiting substantial overshoot. The integrator network was modified by adding capacitor C2 in parallel with the resistor associated with the non-inverting input network. This introduces a zero into the compensation transfer function. The documented form is:

$$V_OUT / V_IN = -(R1/R2) \times (1 + sR2C2) / (1 + sR1C1)$$

The corresponding pole and zero locations are $-1/(R1C1)$ and $-1/(R2C2)$, respectively. The purpose of the added zero is to improve the transient response by reducing overshoot and shortening settling time. The source report does not provide numerical overshoot or settling-time measurements.

5. Control, Measurement, and Range Selection

5.1 Microcontroller

The digital controller is an STM32F103RCT6 from the STM32 family, based on an ARM Cortex-M3 core. The design notes specify a maximum clock frequency of 72 MHz, 256 kB of Flash, and 48 kB of RAM. Its DAC provides the analog reference used by the current-control circuitry, while digital outputs control the range-selection network and relay drivers.

Feature	Documented specification
Core	ARM Cortex-M3, up to 72 MHz
Memory	256 kB Flash; 48 kB RAM
GPIO	37 programmable I/O pins (as listed in the design notes)
Timers	Three 16-bit timers and two 32-bit timers (as documented)
ADC	12-bit, 16 channels
DAC	12-bit, 2 channels
Serial interfaces	3 USART, 2 SPI, 2 I^2C
Other peripherals	RTC, watchdog timer, DMA

An external crystal is used for the microcontroller clock. The design notes specify a 100 nF multilayer decoupling capacitor at each supply pin. A regulated 3.3 V rail is generated using an LM317T, and a USB-to-UART interface is used to communicate with the controller from a host computer.

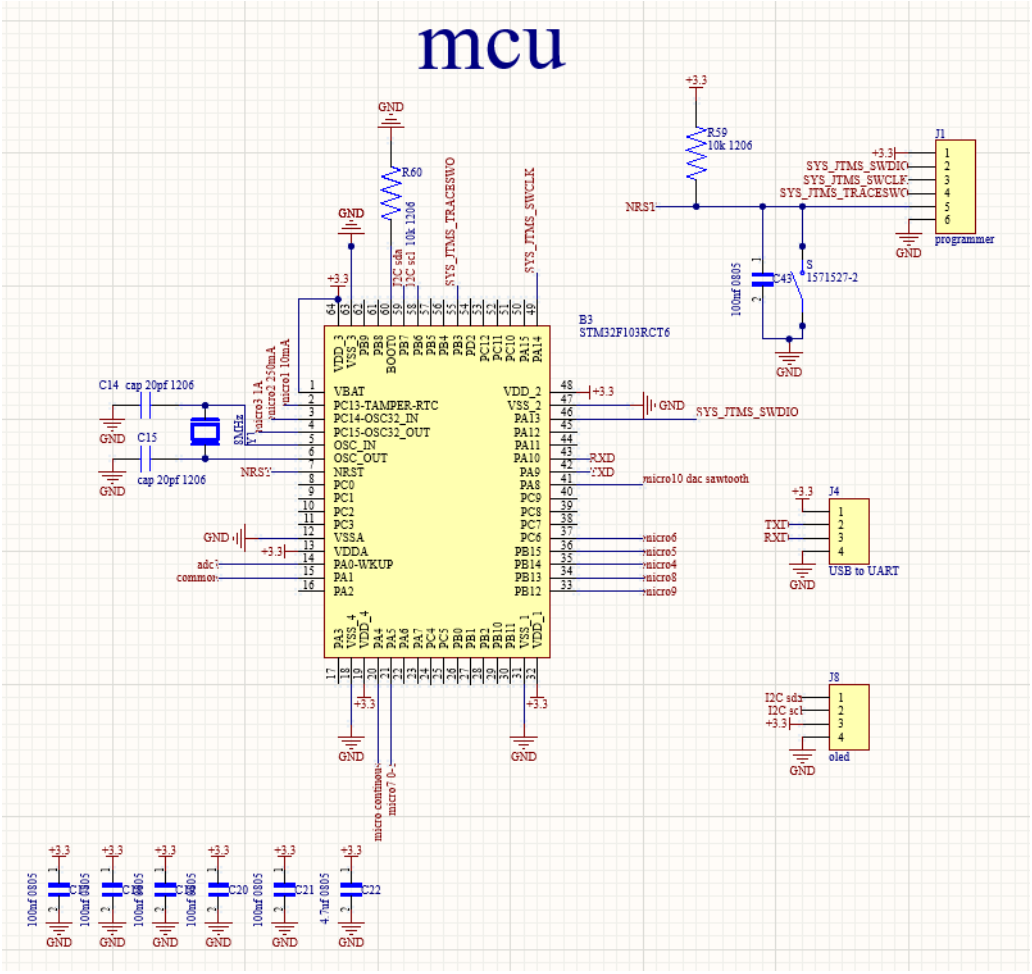


Figure 6. STM32F103RCT6 microcontroller schematic and associated interfaces.

5.2 Relay-Based Path Selection

Because the linear and boost paths are separate, only the selected path should be connected to the DUT at a given time. The design uses two relay types to accommodate the different current levels and contact requirements:

Part	Role	Implementation note
G5NB-1A-E, 12 V	Low-current path selection; contact current rating listed as up to 3 A	BD137 transistor driver
TRA2-L-12VDC-S-H	High-current path selection; contact current rating listed as up to 20 A	BD137 transistor driver
1N5817	Flyback diode across relay coil	Suppresses inductive turn-off transient

The source design notes explain that a single SPDT relay was not considered sufficient for both low- and high-current conditions because the contact minimum-load behavior could affect reliable low-current conduction. Separate relays were therefore selected for the two paths.

5.3 Supply Rails and Interfaces

Rail	Use
±15 V	Dual analog supply for OP07 stages
3.3 V	STM32F103RCT6 supply and DAC reference domain
5 V	74HC4051D multiplexer supply
12 V	Relay coil supply
15 V	TC4428CPA gate-driver supply (as documented)

6. Component Selection

The component choices documented for the two power paths are summarized below. The selected ratings are reproduced from the design notes and should be checked against the exact manufacturer datasheet and operating conditions before procurement or fabrication.

Component	Type	Role in the design
OP07	Precision, low-noise operational amplifier	Used in current regulation and differential/integrator stages; ±15 V supply is specified.
BSP100	N-channel MOSFET	Pass/current-control element in the linear current-sink stage.
74HC4051D	8-channel analog multiplexer/demultiplexer	Selects among feedback networks; three channels are used; 5 V supply.
IRF7303	Dual N-channel MOSFET	Semiconductor switching of the selectable Rsense branches.
BC817	NPN BJT	Drives the MOSFET switching network.
LM317T	Adjustable linear regulator	Provides regulated supply rails, including the 5 V multiplexer rail and 3.3 V controller rail as described.
IRLZ44N	N-channel power MOSFET	Boost-converter switching device; documented VDS rating 55 V and current rating 47 A.
STPS1545FP	Schottky diode	Boost power-path diode; documented 45 V reverse rating and 15 A forward rating.
100 µH toroidal inductor	Power inductor	48 mm outer diameter, 20 A saturation current, 14 mΩ winding resistance, powder-iron material 26.
5 × 47 µF + 100 nF	Capacitor bank	Output filtering and high-frequency bypassing.
ICL8038	Waveform generator	Generates the triangular carrier used for PWM.

Component	Type	Role in the design
TC4428CPA	MOSFET gate driver	Drives the power MOSFET gate from the PWM control signal.
G5NB-1A-E / TRA2-L-12VDC-S-H	Relays	Select the low-current and high-current power paths.
BD137 / 1N5817	Relay driver / flyback diode	Relay coil switching and transient suppression.

> **Component-rating note**
> Ratings stated in the original design record are nominal component-level values. They do not, by themselves, establish safe operating margins in the assembled circuit. Thermal dissipation, MOSFET safe operating area, inductor saturation margin, diode temperature rise, PCB copper capacity, and gate-drive timing require verification in the final design.

7. LTspice Simulation

7.1 Linear Current-Sink Simulation

The continuous current-sink circuit was modeled in LTspice after the topology and component values were selected. The documented example uses Vmicro = 3.1 V and Rsense = 0.31 Ω. The report includes the simulation schematic and an output plot, but it does not provide a numerical table of current error, line regulation, load regulation, or transient response.

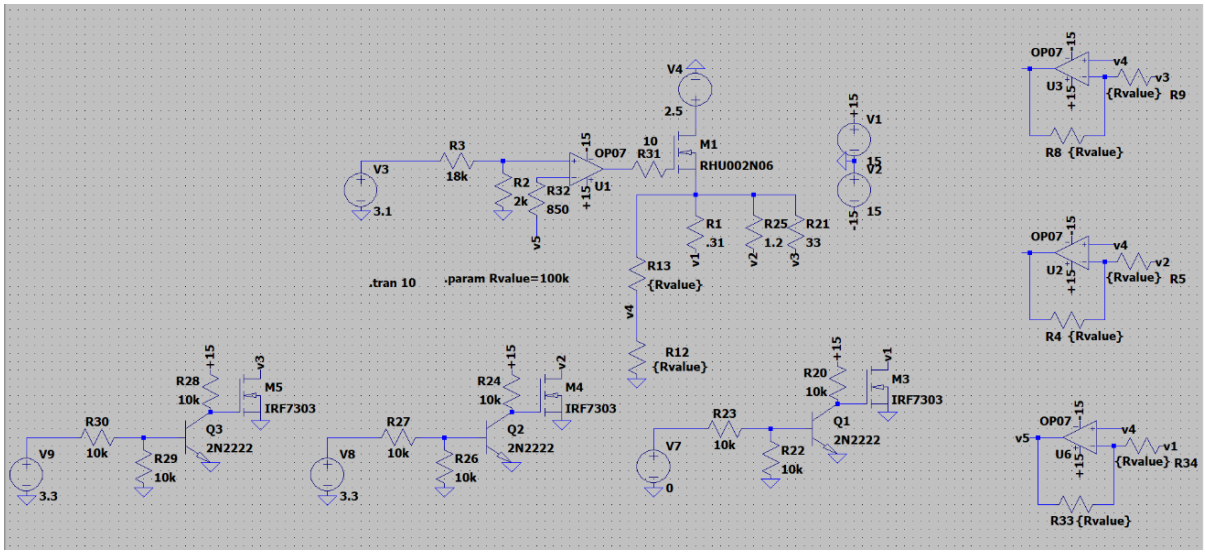


Figure 7. LTspice model of the continuous current-sink stage with selectable sense-resistor branches

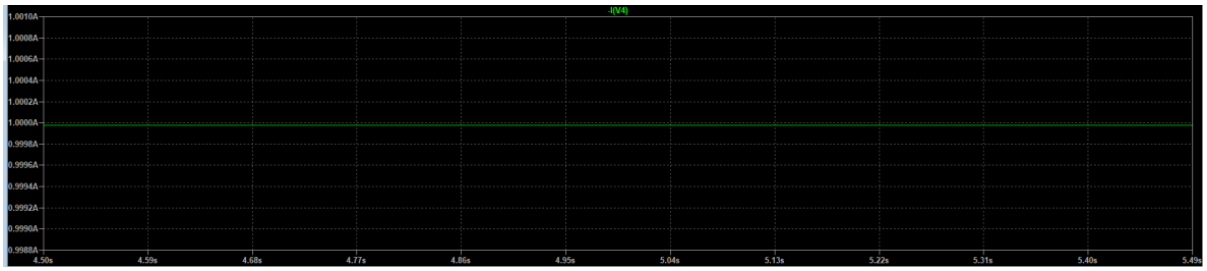


Figure 8. LTspice current waveform for the documented linear-stage simulation.

7.2 Boost Converter Simulation

The boost power stage was first simulated as a converter, and the inductor-current waveform was reviewed. A current-feedback loop was then added, incorporating the shunt measurement, differential amplifier, integrator, PWM comparator, and gate driver. The documented simulation illustrates a change in inductor current from 10 A to 1 A by changing the microcontroller reference voltage.

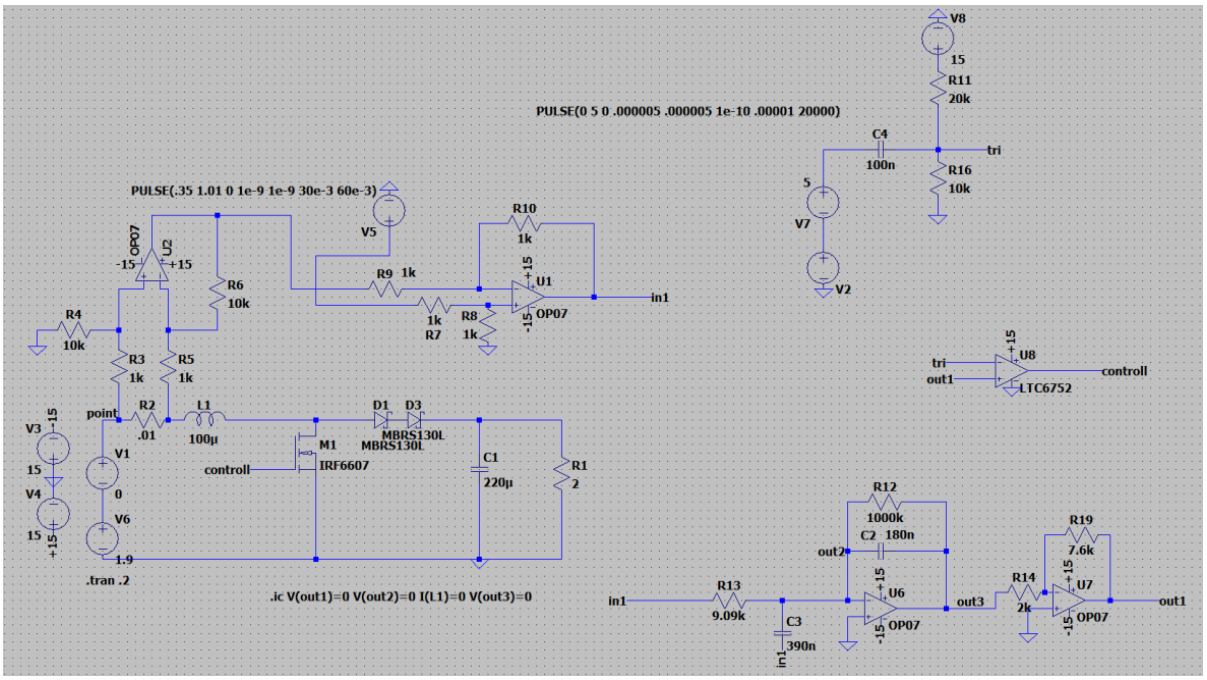


Figure 9. LTspice model of the boost-based current-control loop.

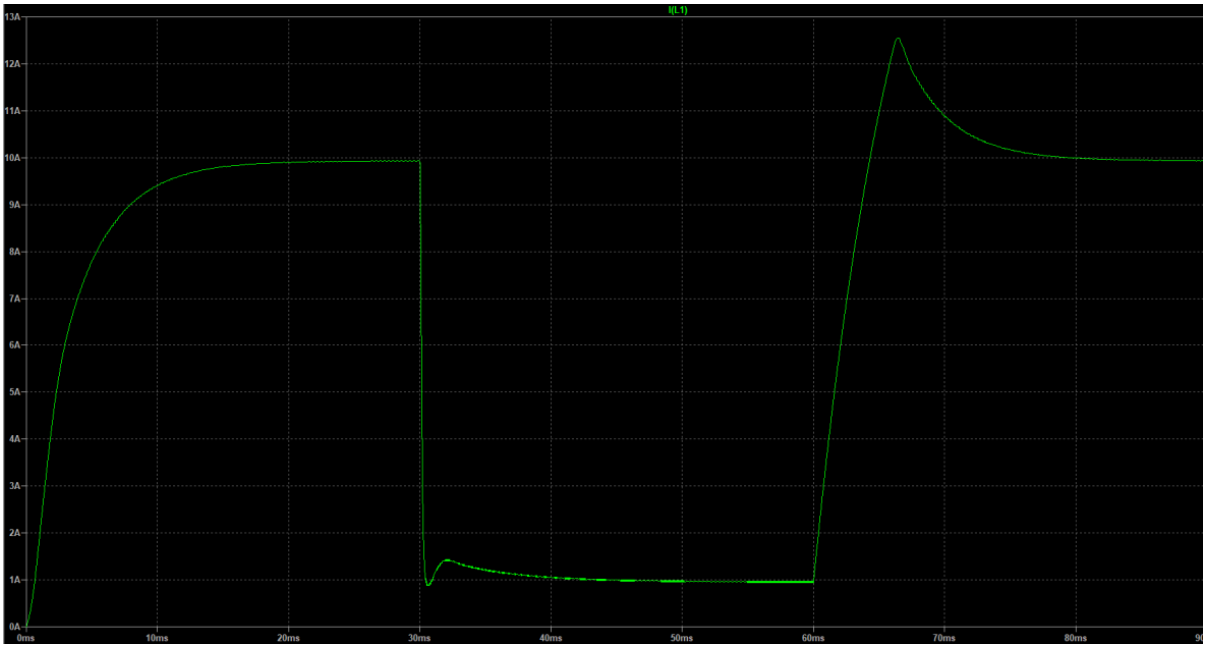


Figure 10. Simulated inductor-current response during transitions between approximately 10 A and 1 A.

7.3 Compensation Study

The initial closed-loop response was described as slow and having significant overshoot. The compensation network was modified to introduce a zero. The report describes the expected direction of improvement—reduced overshoot and shorter settling time—but does not state numerical before/after results.

> **Simulation evidence boundary**
> The available report supports the statement that LTspice simulations were performed and that the current-control response was examined. It does not support claims of measured hardware accuracy, a quantified stability margin, a specific current-regulation percentage, or a verified thermal operating limit.

8. Schematic and PCB Implementation

8.1 Schematic Organization

The complete design was organized into functional schematic blocks in Altium Designer. The documented blocks include the continuous current-sink circuit, boost converter, microcontroller and its peripherals, DUT/IC connection, external supply connections, triangular-wave generator, and voltage-regulator circuitry. Net labels were used to connect the functional blocks.

The schematic documentation includes an overall circuit view and separate sheets for the regulator and waveform-generation section, continuous current-sink stage, boost stage, external supply and IC-pin connections, and microcontroller circuitry.

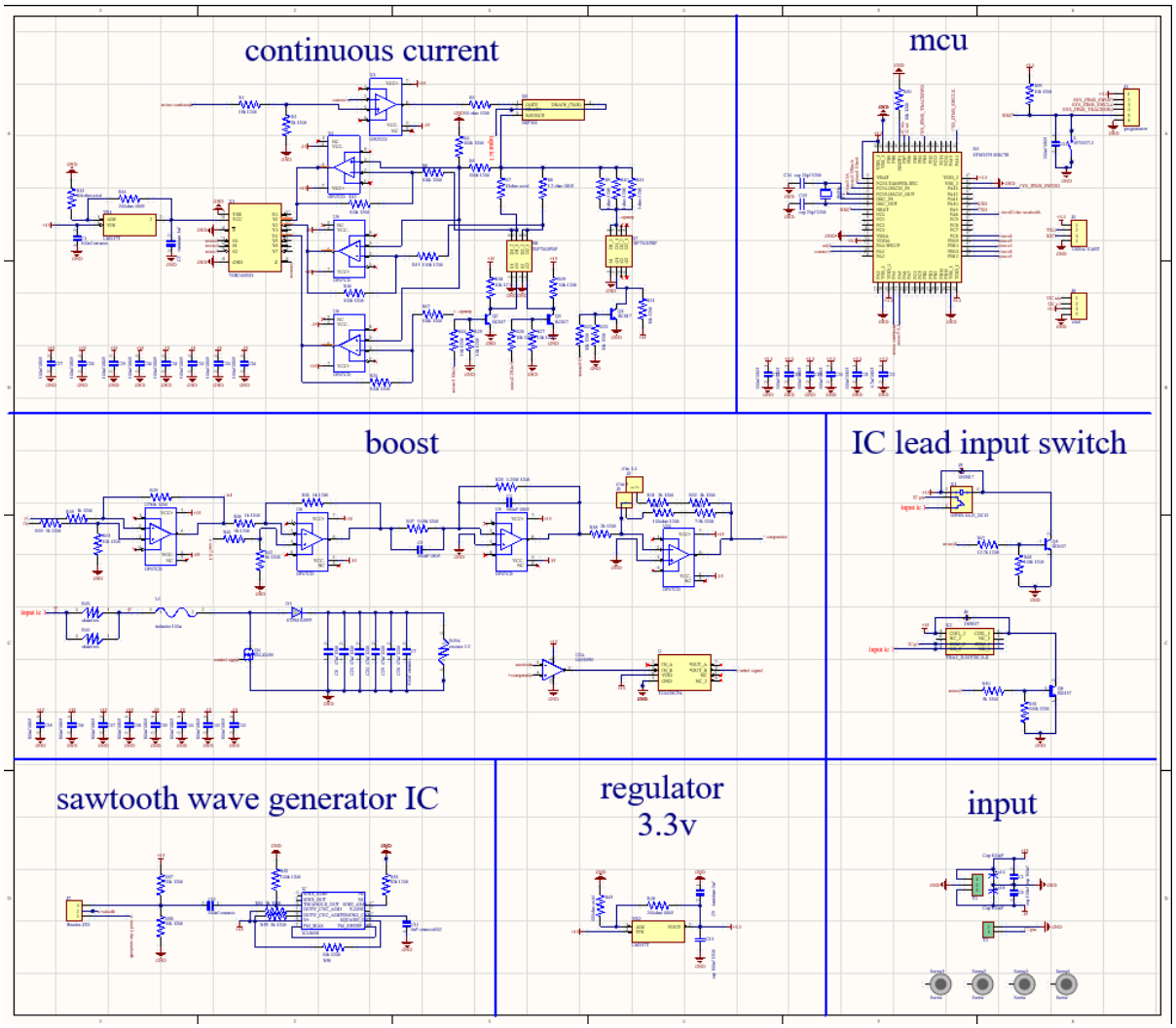


Figure 11. Overall Altium schematic showing the continuous-current, boost, MCU, input-switching, waveform-generator, and regulator sections

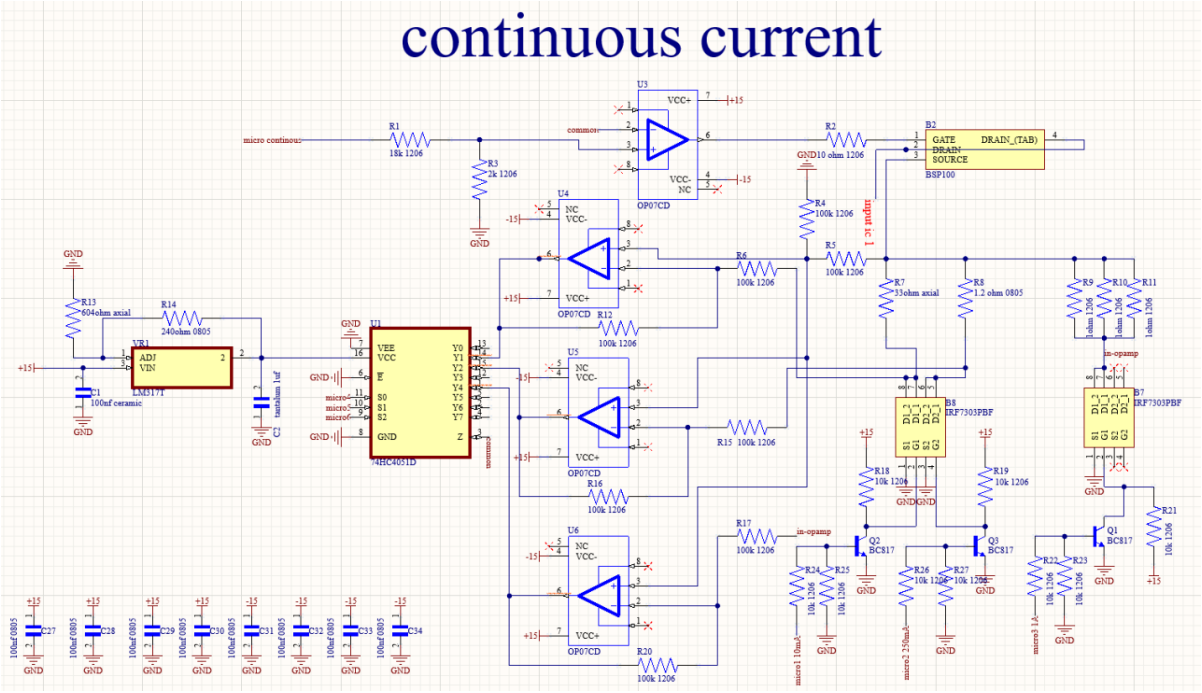


Figure 12. Altium schematic of the continuous-current sink and selectable sensing network.

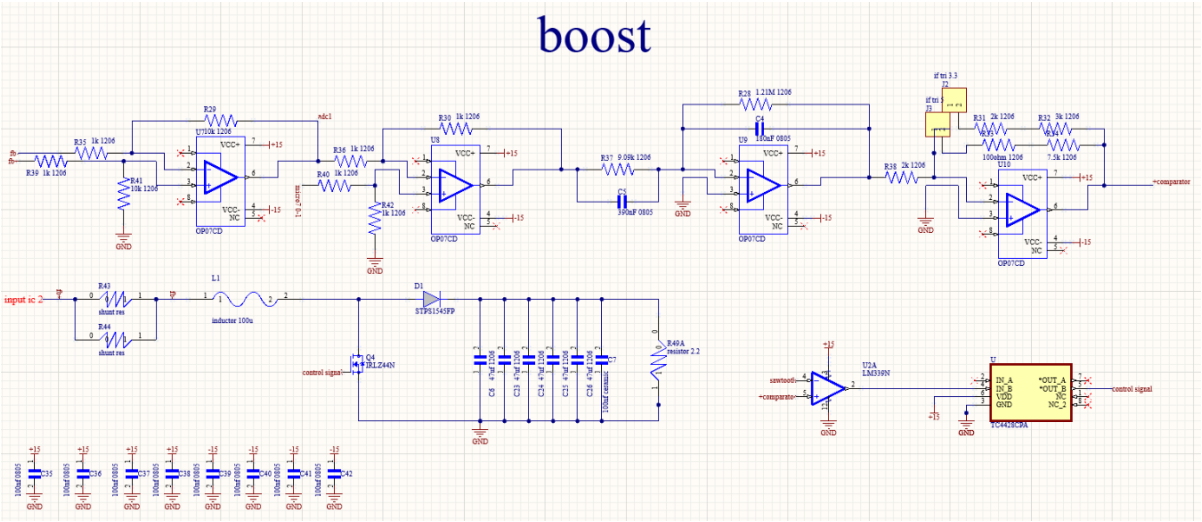


Figure 13. Altium schematic of the boost power stage and associated control circuitry.

8.2 Two-Layer PCB

The PCB is documented as a two-layer design with 35 µm copper. The boost power section is placed on the top side, while the continuous current-sink section is placed on the bottom side. Because the boost path carries high current, wide copper tracks and polygon pours are used to reduce the resistance of the high-current connections.

Ground polygons are placed on both the top and bottom layers. Via stitching connects the two ground regions to form a more continuous ground reference. The report includes top-layer and bottom-layer views, along with 3D views of the front and back of the board.

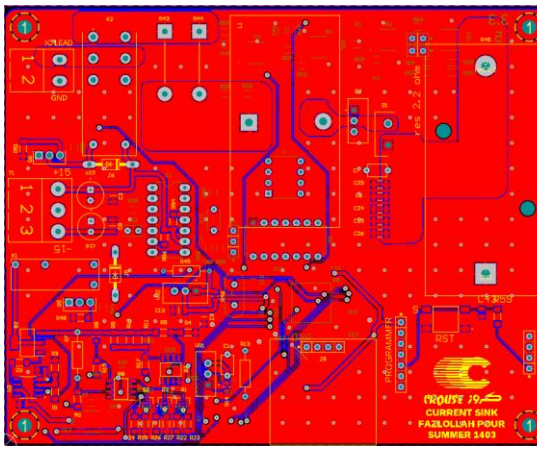


Figure 14a. PCB top-layer layout.

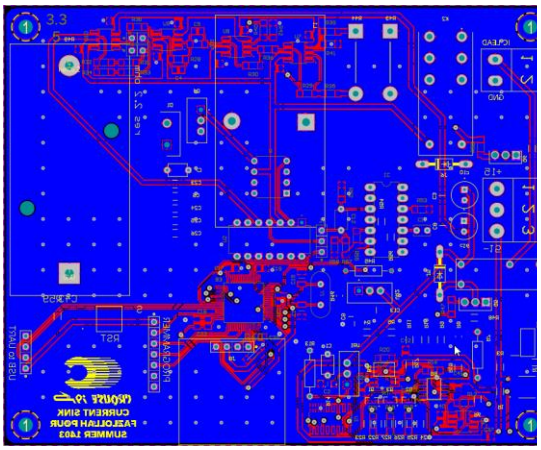


Figure 14b. PCB bottom-layer layout.

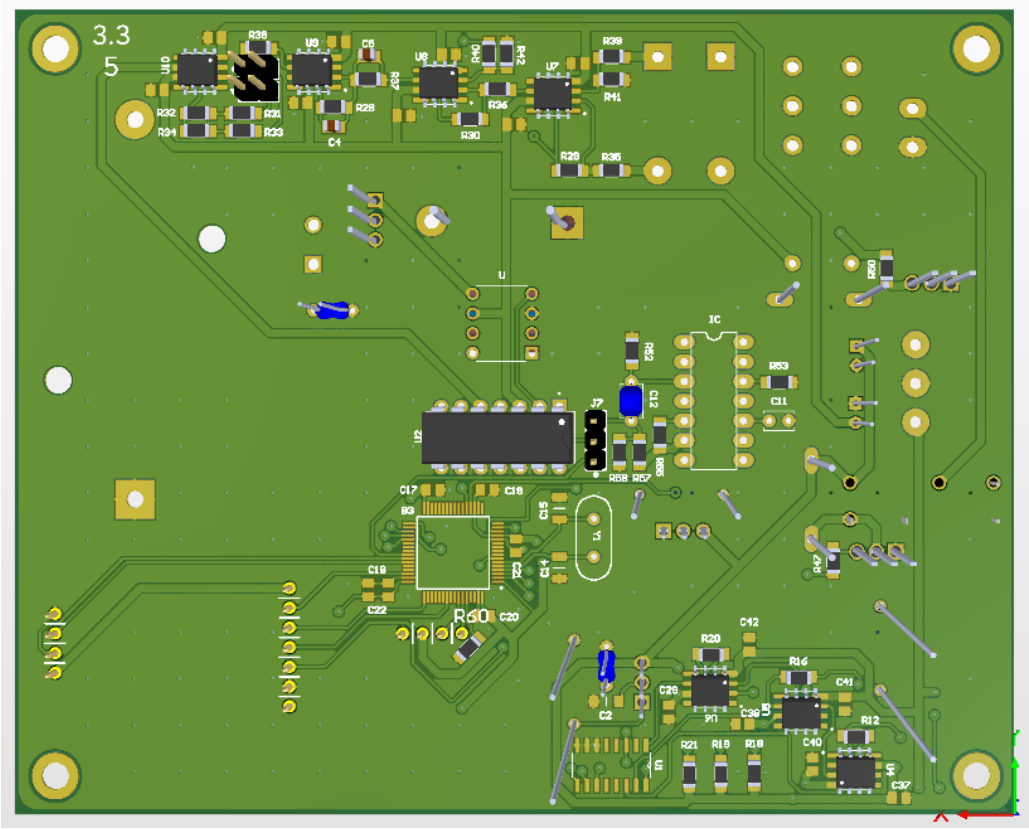


Figure 15. Three-dimensional rear-side view of the PCB layout.

PCB item	Documented implementation
Layer count	2
Copper thickness	35 µm
Top layer	Boost-converter power section
Bottom layer	Continuous current-sink section

PCB item	Documented implementation
High-current routing	Wide tracks and copper polygons
Ground strategy	Top- and bottom-layer GND polygons connected using stitching vias
Design tool	Altium Designer

9. Design Review and Recommended Verification

The design record contains a useful progression from a simple linear current sink to a split-range architecture with a dedicated boost-based high-current stage. Before the design is treated as a fully validated instrument or released as a public engineering reference, the following checks are recommended.

Verification item	Recommended activity
Reconcile current-range calculations	Verify every Rsense value, DAC range, scaling factor, and intended current window against the final schematic. Resolve the inconsistency highlighted in Section 3.4.
Check MOSFET operating limits	Verify VDS, VGS, SOA, dissipation, junction temperature, and transient stress for both the linear pass device and boost switch.
Verify current-sense accuracy	Include shunt tolerance, amplifier offset, resistor-ratio matching, PCB trace resistance, and temperature drift in the error budget.
Validate the boost control loop	Measure or simulate gain/phase margin, overshoot, settling time, inductor-current ripple, and behavior across the full input-voltage and current range.
Check power-stage thermal design	Evaluate inductor copper loss, core loss, diode dissipation, MOSFET switching/conduction loss, and PCB temperature rise.
Verify relay and range switching	Confirm contact ratings, minimum-load behavior, break-before-make sequencing, coil drive, and flyback suppression.
Review PCB high-current layout	Confirm copper width, polygon connections, via current capacity, return paths, and thermal spreading for the maximum intended current.
Perform bench validation	After fabrication, verify current accuracy at multiple setpoints and loads, startup/shutdown behavior, range transitions, protection behavior, and thermal performance.

Recommended publication wording

Until bench measurements are available, describe this work as a circuit-design, LTspice-simulation, and Altium schematic/PCB-layout project. Avoid describing it as a fully tested or calibrated 10 A electronic load.

10. Conclusion

A multi-range voltage-controlled current sink was developed as an active-load concept for the electrical testing and validation of integrated circuits. The design combines a precision op-amp/MOSFET linear current-sink stage for lower and intermediate current levels with a boost-converter-based current-control stage for the high-current region. The architecture uses selectable current-sense elements, differential current measurement, a microcontroller-generated reference, PWM control, and relay-based path selection.

The design process addresses several practical constraints: MOSFET voltage headroom at high current, the influence of relay contact resistance on low-value current-sense resistors, the need for a separate high-current topology, and transient overshoot in the boost feedback loop. LTspice simulations were used to examine the linear and boost-based circuits, while Altium Designer was used to organize the schematic and implement a two-layer PCB layout.

The documented work establishes a detailed design foundation. The next engineering step is to reconcile the current-range calculations with the final schematic, verify component and thermal margins, and complete bench measurements to quantify accuracy, stability, and operating limits.

Appendix A. Key Equations and Nomenclature

A.1 Key Equations

Relationship	Expression
Sense-resistor current	$I \approx V / R_{sense}$
Differential amplifier (unity gain)	$V_{out} = V_2 - V_1$
Differential amplifier (gain 10)	$V_{out} = 10 \times (V_2 - V_1)$
Ideal boost conversion ratio	$V_{out} = V_{in} / (1 - D)$
Approximate boost inductor ripple	$\Delta I_L \approx V_{in} \times D / (L \times f_{sw})$
Modified integrator network	$V_{out}/V_{in} = -(R1/R2) \times (1 + sR2C2)/(1 + sR1C1)$
RC high-pass filter	$H(s) = V_{out}/V_{in} = sRC / (1 + sRC)$

A.2 Nomenclature

Symbol	Definition
D	PWM duty cycle
DUT	Device under test
fsw	Switching frequency
IL	Inductor current
Rsense	Current-sense resistor
VCCS	Voltage-controlled current sink in this project
Vin	Control/reference input voltage, as applicable
Vmicro	Microcontroller DAC output
VDS / VGS	MOSFET drain-to-source / gate-to-source voltage
PWM	Pulse-width modulation

Appendix B. Component Summary

This appendix consolidates the principal components named in the design record. Exact manufacturer part numbers, package variants, and ratings should be confirmed against the final schematic and datasheets.

Subsystem	Principal components
Linear current sink	OP07, BSP100, 74HC4051D, IRF7303, BC817, LM317T, 1% resistors
Boost power stage	IRLZ44N, STPS1545FP, 100 µH toroidal inductor, 5 × 47 µF, 100 nF
Boost control	OP07, ICL8038, TC4428CPA, 10 mΩ current shunt
Digital control	STM32F103RCT6, external crystal, 100 nF decoupling capacitors, USB-to-UART
Path selection	G5NB-1A-E 12 V relay, TRA2-L-12VDC-S-H 12 V relay, BD137, 1N5817